Entanglement Swapping Sandbox – State Based

Instructor Report

Artifact class: Theory, classroom implementation, assessment, and problem companion to the state-vector-based entanglement swapping simulator.

Purpose: Document the physical model, Bell-state measurement construction, concurrence and fidelity interpretation, instructional scope, classroom deployment strategy, and assessment structure supporting the simulator.

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Abstract

This report describes a classroom-friendly, state-vector-based entanglement swapping simulator designed for two-year community colleges and upper-level undergraduate instruction. The simulator assumes familiarity with Dirac notation, normalization, and inner products. Starting from those prerequisites, the report develops entanglement, Bell-state projection, concurrence, and fidelity in the context of a restricted pure-state model. The simulator also includes a simplified pure-state noise treatment through shot-to-shot source perturbations and imperfect Bell-state measurement states. The goal is to preserve conceptual accessibility while making the effect of imperfect preparation and imperfect hub measurement more visible. To facilitate classroom adoption, we provide guided simulator exercises, assessment guidance, and pre- and post-questions, while placing longer derivations in appendices.

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1. Introduction

Entanglement swapping is a central operation in quantum networking. It shows that two qubits can become entangled even when they never interacted directly. The protocol connects three important ideas in quantum information processing: entanglement as a property of a composite state, measurement as an operation that changes the state of a system, and networking as the distribution and transformation of quantum resources across separated locations. In the case of quantum networking, entanglement swapping links symbolic state manipulation to a concrete technological task in the hope of raising interest and awareness of quantum technologies and of reducing entry barriers into quantum behavior for early learners.

This report discusses a state-vector-based approach to entanglement swapping and explains its underlying physics and mathematics. The first part develops the required theory assuming some familiarity with Dirac notation, inner products, normalization, tensor-product basis states, and state-based entanglement measures. The later parts of the report address classroom deployment, guided exploration, assessment, and pre- and post-instruction questions.

The remainder of the report follows that structure closely. Section 2 defines the audience, scope, and instructional purpose. Sections 3 through 8 develop the required state-based theory, beginning with a short refresher and then moving through entanglement, concurrence, fidelity, Bell states, and the state-vector formulation of entanglement swapping. The later sections shift from theory to implementation and classroom use. They explain the design scope of the simulator, outline guided classroom exercises, provide assessment guidance, and conclude with pre-test, post-test, and problem material suitable for direct instructional use. Longer algebraic derivations have been moved to the appendices.

2. Audience, scope, and instructional purpose

This report is written for instructors at two-year community colleges and for instructors teaching upper-level undergraduate students who are early in quantum information science. While the guided exercises, assessment, and hands-on simulator aim at facilitating the translation of quantum concepts from theory into action, the material can also support self-guided learning by quantum-curious students. The intended learners should have some familiarity with kets, inner products, normalization, and the notion of computational basis. The report does not assume prior exposure to density matrices, reduced states, open-system models, or advanced quantum information formalisms.

The scope of the simulator is intentionally narrow. It isolates the principle of entanglement swapping and keeps the mathematical framework at the pure-state level. It therefore excludes density operators, partial traces, mixed states, and full hardware models from the main treatment. This is a deliberate design choice. For the intended audience, the instructional objective is not to approximate every feature of a laboratory platform. The objective is to construct a technically correct pathway from two-qubit states to a branch-conditioned remote entangled state. Once students master that pathway, instructors can connect the same logic to broader topics such as quantum repeaters, teleportation protocols, distributed sensing, and modular quantum architectures.

The simulator includes imperfect source preparation and imperfect Bell-state measurement in a simplified way, but it does so while remaining in a pure-state Monte Carlo picture. Each shot is

treated as one definite pure-state realization, and the notebook reports averages over repeated realizations. This keeps the model accessible for early learners while introducing the idea that entanglement swapping can be sensitive to both resource quality and measurement quality. A more rigorous mixed-state or channel-based treatment is possible and is discussed briefly later in the report.

The simulator is intended for guided exploration during class, with the same instructional sequence also usable for motivated students working independently outside class. The report therefore serves two purposes. First, it gives instructors a technically grounded account of what the simulator computes. Second, it provides a classroom structure through which students can connect notation, equations, and simulator output. A consistent principle throughout the report is that each displayed equation is followed by a direct plain-language explanation of what that equation means physically.

3. A refresher of some quantum fundamentals

For a single qubit, the computational basis states are

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \qquad |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

A general single-qubit pure state is written as

$$|\psi\rangle = a|0\rangle + b|1\rangle,$$

with normalization condition

$$|a|^2 + |b|^2 = 1.$$

This equation states that the total probability associated with the state equals one. The coefficients a and b are probability amplitudes, and the squared magnitudes of those amplitudes determine measurement probabilities in the computational basis.

The inner product $\langle\phi|\psi\rangle$ measures the overlap between two states. In the computational basis,

$$\langle 0|1\rangle = 0, \qquad \langle 1|0\rangle = 0, \qquad \langle 0|0\rangle = 1, \qquad \langle 1|1\rangle = 1,$$

so the inner product between different basis states is zero and we say that the states are orthogonal. The inner product between the same basis states is one, showing that a state with itself has maximum overlap.

For two qubits, the basis states are tensor products of the single-qubit basis vectors:

$$|00\rangle, \quad |01\rangle, \quad |10\rangle, \quad |11\rangle.$$

A general two-qubit pure state is therefore

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

with normalization

$$|a|^2 + |b|^2 + |c|^2 + |d|^2 = 1.$$

4. Entanglement in the pure-state picture

A two-qubit pure state is entangled if it cannot be factored into a product of one state for the first qubit and one state for the second qubit. Product states describe independent single-qubit assignments; entangled states do not admit that separation.

The simulator uses source states of the form

$$|\psi(\alpha, \phi)\rangle = \cos\alpha\,|00\rangle + e^{i\phi}\sin\alpha\,|11\rangle.$$

This source-state family is useful instructionally because each parameter has a clear role. The angle α controls the relative magnitudes of the $|00\rangle$ and $|11\rangle$ amplitudes. The phase ϕ controls the relative phase between those two components.

When $\alpha = 0°$, the state reduces to

$$|\psi(0°, \phi)\rangle = |00\rangle,$$

and when $\alpha = 90°$, the state becomes

$$|\psi(90°, \phi)\rangle = e^{i\phi}|11\rangle.$$

In both limits the state is a product state. The most important point in this state family occurs at $\alpha = 45°$:

$$|\psi(45°, \phi)\rangle = \frac{|00\rangle + e^{i\phi}|11\rangle}{\sqrt{2}}.$$

This equation states that the two basis components have equal magnitude. Within this source-state family, that amplitude balance is the condition for maximal entanglement.

5. Concurrence as a state-based entanglement measure

The simulator reports concurrence as one quantitative measure of entanglement in the swapped remote state. Because the simulator is restricted to pure states, concurrence can be introduced without density operators.

For a general two-qubit pure state

$$|\psi\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

the concurrence is

$$C = 2|ad - bc|.$$

For the simulator, the most relevant case is a state confined to the $|00\rangle$ and $|11\rangle$ sector,

$$|\psi\rangle = a|00\rangle + d|11\rangle.$$

Substituting $b = 0$ and $c = 0$ into the general expression gives

$$C = 2|ad|.$$

This reduced expression is highly instructive. It states that concurrence is zero if either amplitude vanishes and is largest when both amplitudes are present with equal magnitude.

6. Fidelity as a state-closeness measure

The simulator also reports fidelity. Fidelity serves a different purpose from concurrence. Concurrence asks how entangled the remote state is. Fidelity asks how close the remote state is to an intended target state.

In the present simulator, the target state is chosen as the ideal branch-conditioned remote state for the selected source settings and Bell branch, computed with no noise. If the noisy remote state is denoted by $|\psi_{14}^{(\text{noisy})}\rangle$ and the corresponding ideal target by $|\psi_{14}^{(\text{ideal})}\rangle$, then the fidelity is

$$F = \left|\langle\psi_{14}^{(\text{ideal})}|\psi_{14}^{(\text{noisy})}\rangle\right|^2.$$

This quantity lies between 0 and 1. A value near 1 indicates that the noisy remote state remains close to the intended output of the swapping protocol. A smaller value indicates that the noise has moved the branch-conditioned output away from its ideal form.

The pedagogical value of fidelity is that it separates two ideas that students often conflate. A noisy remote state can remain significantly entangled while still deviating from the intended target state. Concurrence and fidelity therefore answer different questions, and that distinction is useful in classroom discussion.

7. Bell states and Bell-state measurement branches

The simulator allows the user to choose a Bell-state measurement branch at the hub. The four Bell states are

$$|\Phi^+\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad |\Phi^-\rangle = \frac{|00\rangle - |11\rangle}{\sqrt{2}},$$
$$|\Psi^+\rangle = \frac{|01\rangle + |10\rangle}{\sqrt{2}}, \qquad |\Psi^-\rangle = \frac{|01\rangle - |10\rangle}{\sqrt{2}}.$$

These four states form an orthonormal basis for two qubits. Measuring in the Bell basis therefore asks a different question from measuring in the computational basis. Instead of asking which computational basis state is present, the Bell-state measurement asks which entangled two-qubit pattern is present.

This distinction is important in entanglement swapping. The protocol does not apply a standard computational-basis measurement to the middle qubits. It applies a Bell-basis projection. The outcome selects a specific branch, and that branch determines the resulting state of the two remote qubits.

8. State-vector formulation of entanglement swapping

The protocol uses four qubits. Qubits 1 and 2 form source pair A, and qubits 3 and 4 form source pair B. The input state is

$$|\Psi\rangle_{1234} = |\psi_A\rangle_{12} \otimes |\psi_B\rangle_{34}.$$

The simulator defines the source states as

$$|\psi_A\rangle = \cos\alpha_A\,|00\rangle + e^{i\phi_A}\sin\alpha_A\,|11\rangle,$$

$$|\psi_B\rangle = \cos\alpha_B\,|00\rangle + e^{i\phi_B}\sin\alpha_B\,|11\rangle.$$

The middle qubits, 2 and 3, are then projected onto a chosen Bell branch. If the selected branch is $|B\rangle_{23}$, the remote state of qubits 1 and 4 is defined by

$$|\chi\rangle_{14} \propto {}_{23}\langle B|\Psi\rangle_{1234}.$$

This is the central equation of the simulator. The Bell bra acts on the middle qubits only. The projection removes qubits 2 and 3 from the explicit state expression and leaves a new two-qubit state on qubits 1 and 4.

The branch probability is the norm squared of the unnormalized remote state,

$$p_{\text{branch}} = \langle\chi|\chi\rangle.$$

The normalized remote state is then

$$|\psi_{14}\rangle = \frac{|\chi\rangle_{14}}{\sqrt{p_{\text{branch}}}}.$$

Once normalization is complete, the simulator computes concurrence and fidelity for $|\psi_{14}\rangle$.

9. What the simulator computes for the chosen source family

Because both source states are restricted to the $|00\rangle$ and $|11\rangle$ sector, the simulator exhibits a simple structural selection rule. In the ideal model, the even Bell branches $|\Phi^+\rangle$ and $|\Phi^-\rangle$ are compatible with the input structure, whereas the odd branches $|\Psi^+\rangle$ and $|\Psi^-\rangle$ are ideally suppressed. This result follows directly from basis compatibility.

For the even branches, the swapped remote state typically remains in the $|00\rangle$ and $|11\rangle$ sector. In that regime, concurrence reduces to the simple form $C = 2|ad|$. The simulator therefore shows high concurrence when the remote amplitudes are balanced and low concurrence when one amplitude dominates.

Fidelity adds a second layer of interpretation. A remote state can remain moderately or even strongly entangled while no longer matching the intended ideal swapped state closely. For instruction, this is useful because it helps separate resource quality from target accuracy.

10. Role of phase in this simulator

One might expect that changing phase must always increase or decrease the entanglement. The simulator helps refine that idea. The source phases ϕ_A and ϕ_B change the relative phase between the basis components in each source state, so a phase change does change the state itself. However, in the ideal source-state family used here, the swapped remote state for the even Bell branches remains in the $|00\rangle/|11\rangle$ sector. For a normalized state of the form

$$|\psi\rangle = a|00\rangle + d|11\rangle,$$

the concurrence is

$$C = 2|ad|.$$

This expression depends on the magnitudes of the two amplitudes and not on their relative phase. In this simulator, source phases can change the symbolic form of the remote state without necessarily changing concurrence.

Fidelity, however, can respond to these state changes because it measures overlap with a target state, not only entanglement strength. This gives instructors a natural way to explain why two metrics may respond differently to the same perturbation.

11. Noise model and its interpretation

The simulator includes a pedagogical noise model that remains entirely within a pure-state treatment. Instead of introducing density matrices or quantum channels, the simulator treats each shot as one definite pure-state realization with perturbed preparation and measurement settings. The reported noisy metrics are averages over repeated shots.

The noise model contains three layers.

First, the simulator includes shot-to-shot parameter jitter in the source settings. In that case the source states are prepared as

$$|\psi_A^{(\text{shot})}\rangle = \cos(\alpha_A + \delta\alpha_A)\,|00\rangle + e^{i(\phi_A+\delta\phi_A)}\sin(\alpha_A + \delta\alpha_A)\,|11\rangle,$$

$$|\psi_B^{(\text{shot})}\rangle = \cos(\alpha_B + \delta\alpha_B)\,|00\rangle + e^{i(\phi_B+\delta\phi_B)}\sin(\alpha_B + \delta\alpha_B)\,|11\rangle.$$

These terms model imperfect control over the intended preparation parameters.

Second, the simulator allows additional source-state perturbations within the same $|00\rangle/|11\rangle$ sector. In practical terms, the amplitudes of the two source components are perturbed and then renormalized. This acts as a simple effective dial for imperfect source quality while keeping the state-vector picture intact.

Third, the simulator allows perturbations of the Bell-state measurement itself. Instead of projecting onto an exact Bell state in every shot, the simulator can project onto a perturbed Bell-like state and then renormalize. This provides a simple effective dial for imperfect hub measurement or imperfect Bell-state discrimination.

Together, these additions make two instructional points more visible. First, imperfect source states can strongly degrade the swapped output. Second, imperfect Bell-state measurement can also degrade the output, even when the source settings are otherwise favorable.

> **Interpretation of the noise model.** The simulator should be described as a *pure-state Monte Carlo teaching model with effective source and BSM perturbations*. Each shot remains a pure-state realization. The notebook averages the resulting branch probabilities, concurrences, and fidelities over many shots.

This wording is important because it prevents overstatement. The simulator is not intended to be a full open-system hardware model. Instead, it offers an accessible way to show that entanglement swapping can be sensitive to imperfect preparation and imperfect hub operations without requiring a density-matrix formalism at the introductory stage.

12. What a more rigorous treatment would require

A more rigorous treatment of noise would naturally move beyond the pure-state Monte Carlo picture. For example, partial dephasing, depolarization, detector inefficiency, or incoherent branch mixing are more naturally represented with density operators and quantum channels. In that language, one would propagate a density matrix rather than a single ket and would treat the Bell-state measurement as an operation acting on the density operator.

This more advanced treatment is important for realistic hardware analysis, but it is not necessary for the instructional goals of the present simulator. The simplified model keeps the mathematics at a level accessible to early learners while still giving instructors a defensible way to discuss imperfect state preparation and imperfect hub measurement. The simulator can therefore be viewed as a bridge: it teaches the basic protocol in a transparent state-vector language, while also motivating why later coursework may need mixed states and open-system tools.

13. Broader significance and outlook

Although the simulator is framed as a quantum networking tool, the underlying ideas extend beyond networking. Entanglement swapping is directly relevant to quantum repeaters, where intermediate stations are used to distribute entanglement across longer distances. The same branch-conditioned logic also appears in teleportation-style protocols, distributed sensing schemes, and modular architectures for quantum information processing.

This broader context should appear in instruction because it helps students identify the simulator as a model of a general quantum-information pattern rather than as a single isolated example. Even when future hardware platforms differ in their implementation details, the state-based logic of preparation, projection, conditioning, and entanglement as a resource remains relevant.

14. Suggested classroom implementation

The recommended instructional progression is sequential. Begin with the source-state family. Ask students to identify what changes when α varies and what changes when ϕ varies. Then move to

concurrence in the restricted $|00\rangle/|11\rangle$ sector so that students understand why balanced amplitudes produce maximal entanglement. Only after those ideas are secure should the instructor introduce the four-qubit swapping protocol and the Bell-state measurement branches.

After students understand the ideal protocol, introduce fidelity as a second reported metric. This is a good moment to emphasize that entanglement strength and state closeness are not the same quantity. Then introduce noise in stages: first parameter jitter, then source-quality perturbation, then BSM perturbation. This order preserves conceptual control and helps students distinguish between imperfect preparation and imperfect hub measurement.

15. Guided classroom exercises

Exercise 1: analyzing the source-state family

Set a single source state to $\alpha = 0°$, then to $45°$, and then to $90°$, while keeping the phase fixed. Ask students to write the state explicitly in each case and to state whether the state is entangled. The intended conclusion is that the source-state family moves from $|00\rangle$ to a balanced Bell-like superposition and then to $|11\rangle$.

Exercise 2: interpreting concurrence quantitatively

Hold the state in the $|00\rangle/|11\rangle$ sector and write it as

$$|\psi\rangle = a|00\rangle + d|11\rangle.$$

Ask students to compute concurrence using

$$C = 2|ad|.$$

Then give them several simple amplitude pairs. The objective is to make students interpret concurrence numerically rather than treating it as a hidden software output.

Exercise 3: following the entanglement swapping step

Write the four-qubit input state as

$$|\Psi\rangle_{1234} = |\psi_A\rangle_{12} \otimes |\psi_B\rangle_{34}.$$

Then write the branch-selected remote state as

$$|\chi\rangle_{14} \propto {}_{23}\langle B|\Psi\rangle_{1234}.$$

Ask students to explain what this projection does. The target explanation is that the Bell-state measurement acts on qubits 2 and 3 and leaves a new state on qubits 1 and 4.

Exercise 4: branch compatibility

Fix the source states and compare the four Bell branches. Ask students which branches carry nonzero probability and why. The expected explanation is that the chosen source-state family contains only $|00\rangle$ and $|11\rangle$ basis support, so even Bell branches are compatible with the source structure and odd Bell branches are ideally suppressed in the unperturbed model.

Exercise 5: comparing concurrence and fidelity

Set both sources to balanced amplitudes and vary the phases ϕ_A and ϕ_B. Ask students to record the symbolic form of the remote state, the reported concurrence, and the reported fidelity. The central question is: can the state change without the concurrence changing by the same amount? Students should conclude that concurrence and fidelity track different properties of the remote state.

Exercise 6: robustness under imperfect source preparation

Introduce nonzero source perturbation while keeping BSM perturbation zero. Ask students to compare ideal and noisy averages. Their explanation should identify the new source-noise dial as an effective model of imperfect state preparation within a pure-state treatment.

Exercise 7: robustness under imperfect Bell-state measurement

Keep the sources ideal and introduce nonzero BSM perturbation. Ask students to compare branch probability, concurrence, and fidelity with the ideal case. The target explanation is that imperfect hub measurement can degrade the swapped output even when the initial sources are well prepared.

16. Assessment guidance

Assessment should emphasize interpretation, not algebraic length. Students should demonstrate that they can read a state, identify its basis structure, infer whether branch compatibility is present, explain what concurrence measures, and explain what fidelity measures.

A useful assessment strategy is to score student work in four dimensions. First, evaluate whether the student identifies the relevant state components correctly. Second, evaluate whether the student explains what a displayed equation means physically. Third, evaluate whether the student connects simulator output to input-state structure. Fourth, evaluate whether the student distinguishes between entanglement strength and state closeness.

17. Pre-test questions

The following pre-test questions are intended as short conceptual prompts given before the simulator activity.

Pre-test 1: What does it mean for a two-qubit state to be entangled? Answer in words.

Pre-test 2: Consider the state $|00\rangle + |11\rangle$. What additional step is required before this is a valid quantum state, and why?

Pre-test 3: Explain the difference between the basis states $|01\rangle$ and $|10\rangle$.

Pre-test 4: If a two-qubit state contains only one basis component, such as $|11\rangle$, would you expect it to be entangled? Explain your reasoning.

Pre-test 5: What information does an inner product provide about two quantum states?

Pre-test 6: What do you think a measurement on part of a multi-qubit system can do to the state of the remaining qubits?

18. Post-test questions

Post-test 1: Consider the source-state family

$$|\psi(\alpha, \phi)\rangle = \cos\alpha\,|00\rangle + e^{i\phi}\sin\alpha\,|11\rangle.$$

Explain what changing α does to the state and what changing ϕ does to the state.

Post-test 2: Why does the state with $\alpha = 45°$ have maximal entanglement within this source-state family?

Post-test 3: In the simulator, what does the equation

$$|\chi\rangle_{14} \propto {}_{23}\langle B|\Psi\rangle_{1234}$$

mean physically?

Post-test 4: Why do the odd Bell branches ideally vanish for the chosen source-state family?

Post-test 5: For a state of the form

$$|\psi\rangle = a|00\rangle + d|11\rangle,$$

what does the concurrence formula

$$C = 2|ad|$$

measure in words?

Post-test 6: What is the difference between concurrence and fidelity in this simulator?

Post-test 7: Why is the noise model still described as a pure-state model even though it includes imperfect source preparation and imperfect Bell-state measurement?

19. Sample instructor commentary for equation-centered teaching

Because the report emphasizes active explanation of equations, instructors may find it useful to adopt a consistent verbal pattern in class. After writing an equation, state explicitly what the equation does. For example, after writing

$$|\Psi\rangle_{1234} = |\psi_A\rangle_{12} \otimes |\psi_B\rangle_{34},$$

one can say: "This equation builds the full four-qubit input state from the two source pairs." After writing

$$|\chi\rangle_{14} \propto {}_{23}\langle B|\Psi\rangle_{1234},$$

one can say: "This equation projects the middle qubits onto the selected Bell branch and leaves a remote state on qubits 1 and 4." After writing

$$C = 2|ad|,$$

one can say: "This equation measures how strongly the two surviving basis amplitudes coexist in the remote state." After writing

$$F = \left|\langle\psi_{14}^{(\text{ideal})}|\psi_{14}^{(\text{noisy})}\rangle\right|^2,$$

one can say: "This equation measures how close the noisy remote state remains to the intended ideal swapped state."

20. Concluding remarks

The state-vector-based entanglement swapping simulator provides a technically controlled way to introduce an important quantum networking protocol to early quantum learners. Its main strength lies in the clarity of its mathematical structure. The source states are transparent, the Bell-state projection can be expressed with an inner product, the branch probability is a norm, and the final entanglement measures can be computed directly from amplitudes and state overlap in the restricted pure-state setting.

The simulator is especially useful because it combines conceptual clarity with visible control dials for imperfect source preparation and imperfect Bell-state measurement while remaining mathematically accessible. Framed in that way, it can serve not only as an introduction to quantum networking but also as a reusable instructional platform for state preparation, entanglement measures, projection-based measurement, branch-conditioned outcomes, and the distinction between simplified pure-state noise models and more advanced mixed-state treatments.

21. Problems

Problem 1: normalization of a Bell-like source state

Consider the state

$$|\psi\rangle = \cos\alpha\,|00\rangle + e^{i\phi}\sin\alpha\,|11\rangle.$$

Show explicitly that this state is normalized for any choice of α and ϕ.

Problem 2: identifying product-state limits

Evaluate the source-state family at $\alpha = 0°$ and $\alpha = 90°$. Write the resulting states explicitly and explain why neither state is entangled.

Problem 3: concurrence in the restricted sector

For each of the following states, compute the concurrence and state whether the state is entangled:

$$|\psi_1\rangle = |00\rangle, \qquad |\psi_2\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}, \qquad |\psi_3\rangle = \frac{\sqrt{3}}{2}|00\rangle + \frac{1}{2}|11\rangle.$$

Problem 4: inner product of Bell states

Are $|\Phi^+\rangle$ and $|\Phi^-\rangle$ orthogonal? Explain.

Problem 5: reading the Bell projection equation

In your own words, explain the physical meaning of

$$|\chi\rangle_{14} \propto {}_{23}\langle B|\Psi\rangle_{1234}.$$

Your explanation should identify which qubits are measured, which qubits remain, and why the resulting state is called branch-conditioned.

Problem 6: branch compatibility

The source states in the simulator contain only $|00\rangle$ and $|11\rangle$ components. Use that fact to explain why the odd Bell branches $|\Psi^+\rangle$ and $|\Psi^-\rangle$ are ideally absent in the unperturbed model.

Problem 7: interpreting branch probability

The simulator reports a branch probability

$$p_{\text{branch}} = \langle\chi|\chi\rangle.$$

Explain what this number means physically. Then explain why the state must still be normalized after the branch has been selected.

Problem 8: fidelity versus concurrence

A swapped remote state has high concurrence but fidelity below one. Explain how that can happen and what each metric is telling you.

Problem 9: noise interpretation

The simulator includes parameter jitter, source perturbation, and BSM perturbation. Explain what kinds of imperfections these are intended to model and why the simulator can still be described as a pure-state Monte Carlo model.

Problem 10: broader impact

Write a short paragraph explaining why entanglement swapping is useful beyond a single classroom example. Your answer should mention at least one of the following: quantum repeaters, teleportation-

style protocols, distributed sensing, or modular quantum architectures.

A. Worked example: projection onto the $|\Phi^+\rangle_{23}$ branch

This appendix records the explicit projection for the ideal source family and the Φ^+ branch.

We choose

$$|\psi_A\rangle_{12} = \cos\alpha_A\,|00\rangle_{12} + e^{i\phi_A}\sin\alpha_A\,|11\rangle_{12},$$
$$|\psi_B\rangle_{34} = \cos\alpha_B\,|00\rangle_{34} + e^{i\phi_B}\sin\alpha_B\,|11\rangle_{34},$$

and

$$|\Phi^+\rangle_{23} = \frac{|00\rangle_{23} + |11\rangle_{23}}{\sqrt{2}}.$$

The full four-qubit input state is

$$|\Psi\rangle_{1234} = |\psi_A\rangle_{12} \otimes |\psi_B\rangle_{34}.$$

Expanding gives

$$\begin{aligned}|\Psi\rangle_{1234} = {} & \cos\alpha_A\cos\alpha_B\,|0000\rangle \\ & + \cos\alpha_A e^{i\phi_B}\sin\alpha_B\,|0011\rangle \\ & + e^{i\phi_A}\sin\alpha_A\cos\alpha_B\,|1100\rangle \\ & + e^{i(\phi_A+\phi_B)}\sin\alpha_A\sin\alpha_B\,|1111\rangle.\end{aligned}$$

Projecting qubits 2 and 3 onto Φ^+ leaves the unnormalized remote state

$$|\chi\rangle_{14} \propto \frac{1}{\sqrt{2}}\left(\cos\alpha_A\cos\alpha_B\,|00\rangle_{14} + e^{i(\phi_A+\phi_B)}\sin\alpha_A\sin\alpha_B\,|11\rangle_{14}\right).$$

The branch probability is

$$p_{\Phi^+} = \frac{1}{2}\left(\cos^2\alpha_A\cos^2\alpha_B + \sin^2\alpha_A\sin^2\alpha_B\right).$$

The normalized remote state is then

$$|\psi_{14}\rangle = \frac{\cos\alpha_A\cos\alpha_B\,|00\rangle_{14} + e^{i(\phi_A+\phi_B)}\sin\alpha_A\sin\alpha_B\,|11\rangle_{14}}{\sqrt{\cos^2\alpha_A\cos^2\alpha_B + \sin^2\alpha_A\sin^2\alpha_B}}.$$

In this ideal source family, the remote state remains in the $|00\rangle_{14}/|11\rangle_{14}$ sector, which is why concurrence reduces to $C = 2|ad|$.

B. How the pure-state noise model is implemented

The simulator treats each shot as a pure-state realization.

For source perturbation, the simulator begins with the intended restricted-sector source state and perturbs the relevant amplitudes and phases, then renormalizes. Symbolically, one may think of a noisy source shot as

$$|\widetilde{\psi}\rangle \propto (\,a + \delta a\,)|00\rangle + (\,d + \delta d\,)|11\rangle,$$

followed by normalization. Here the perturbations δa and δd represent effective source imperfections rather than a full microscopic hardware model.

For BSM perturbation, the selected Bell-state projector is replaced by a perturbed Bell-like state,

$$|\widetilde{B}\rangle \propto |B\rangle + \text{small admixture of other Bell branches},$$

and the projection step becomes

$$|\widetilde{\chi}\rangle_{14} \propto {}_{23}\langle\widetilde{B}|\Psi\rangle_{1234}.$$

Again, this is an effective pure-state perturbation model, not a mixed-state measurement channel.

C. Extension to a source family that can produce phase-dependent concurrence

The main text studies a restricted source-state family in which each source pair occupies only the $|00\rangle/|11\rangle$ sector. That restriction keeps the swapped remote state in the same sector for the even Bell branches and leads to a concurrence of the form $C = 2|ad|$. In that setting, the source phases can change the symbolic form of the remote state without necessarily changing concurrence.

When the source-state family is enlarged so that the swapped remote state contains all four computational-basis components,

$$|\psi_{14}\rangle = a|00\rangle + b|01\rangle + c|10\rangle + d|11\rangle,$$

the concurrence becomes

$$C = 2|ad - bc|.$$

In that more general case, different amplitude products can interfere, and relative phase can directly affect concurrence. This appendix-level observation is useful for instructors because it clarifies that the phase-independence seen in the main simulator is a consequence of the chosen restricted source family, not a universal feature of entanglement swapping.